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Image forming apparatus

Abstract

An image forming apparatus includes a casing including a discharge port, an upper unit arranged above the casing, a first supporting member supporting the upper unit, and a second supporting member supporting the upper unit. The casing includes a first side plate, a second side plate, and a frame member arranged downstream of the discharge port in the sheet discharge direction. In the sheet width direction, the first side plate and the first supporting member are arranged on a first side, and the second side plate and the second supporting member are arranged on a second side opposite to the first side. The frame member includes a first surface connected to the first side plate, a second surface connected to the second side plate, a third surface connected to the first supporting member, and a fourth surface connected to the second supporting member.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) This disclosure relates to an image forming apparatus that forms an image on a recording material.

Description of the Related Art

(2) In Japanese Patent Application Laid-Open No. 2019-101077, it is described that, by securing a conveyance guide with respect to both a casing of the image forming apparatus and an optional apparatus secured to the top of the casing, the rigidity of an image forming apparatus is enhanced.

(3) In the abovementioned literature, the conveyance guide is disposed upstream of a discharge port in a sheet discharge direction in which the sheet is discharged from the discharge port of the casing.

SUMMARY OF THE INVENTION

(4) The present invention provides an image forming apparatus that can increase rigidity in its part downstream of a discharge port in a sheet discharge direction.

(5) According to an aspect of the invention, an image forming apparatus includes a casing configured to accommodate an image forming portion that is configured to form an image on a sheet, the casing including a discharge port through which the sheet on which the image has been formed by the image forming portion is discharged in a sheet discharge direction, an upper unit arranged above the casing, a first supporting member supporting the upper unit, and a second supporting member supporting the upper unit, wherein the casing includes a first side plate extending in a direction intersecting with a sheet width direction perpendicular to the sheet discharge direction, a second side plate extending in a direction intersecting with the sheet width direction, and a frame member arranged downstream of the discharge port in the sheet discharge direction and configured to connect the first side plate and the second side plate, wherein, in the sheet width direction, the first side plate and the first supporting member are arranged on a first side with respect to a center of the casing, and the second side plate and the second supporting member are arranged on a second side opposite to the first side with respect to the center of the casing, and wherein the frame member includes a first surface connected to the first side plate, a second surface connected to the second side plate, a third surface connected to the first supporting member, and a fourth surface connected to the second supporting member.
(6) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram illustrating an image forming apparatus of a first embodiment.
(2) FIG. **2** is a perspective view illustrating part of a frame of the image forming apparatus of the first embodiment.
(3) FIG. **3** is a perspective view illustrating part of the frame of the image forming apparatus of the first embodiment.
(4) FIG. **4** is a perspective view illustrating a frame of an image forming apparatus of a variant example.
(5) FIG. **5** is a schematic diagram illustrating the image forming apparatus of the variant example.
(6) FIG. **6** is a schematic diagram illustrating a positional relationship between a sheet discharge tray and a front upper stay of the first embodiment.
(7) FIG. **7**A is a schematic diagram illustrating a left side plate of the variant example.
(8) FIG. **7**B is a schematic diagram illustrating a left side plate of the first embodiment.
(9) FIG. **8** is a perspective view illustrating part of the frame of the image forming apparatus of the first embodiment.
(10) FIG. **9** is a perspective view illustrating part of the frame of the image forming apparatus of the first embodiment.
(11) FIG. **10** is a perspective view illustrating a frame of a sheet processing apparatus of the first embodiment.
(12) FIG. **11** is a perspective view illustrating the frame of the sheet processing apparatus of the first embodiment.
(13) FIG. **12** is a schematic diagram illustrating a connection configuration between the sheet processing apparatus and a document reading apparatus of the first embodiment.
(14) FIG. **13** is a perspective view illustrating part of the frame of the image forming apparatus of the first embodiment.
(15) FIG. **14** is a perspective view illustrating the frame of the image forming apparatus of the first embodiment.
(16) FIG. **15** is a perspective view illustrating the frame of the image forming apparatus of the first

embodiment.
(17) FIG. **16** is a perspective view illustrating part of the frame of the image forming apparatus of the first embodiment.
(18) FIG. **17** is a perspective view illustrating a frame of an image forming apparatus of a second embodiment.

DESCRIPTION OF THE EMBODIMENTS

(19) Hereinafter, with reference to drawings, embodiments of the present disclosure will be described.
(20) In the following description and drawings, unless specifically stated otherwise, the term “down” or “below” refers to the direction of gravity when an image forming apparatus is installed on a horizontal plane, and “up” or “above” refers to the direction opposite to the direction of gravity. In a sheet discharge direction D**1** in which a sheet is discharged from a discharge port **100***d* (refer to FIG. **1**) toward a sheet discharge tray **135** of a printer body **100**, a downstream side is referred to as “front” or “front side”, and an upstream side is referred to as “back” or “back side”. The right side when viewing the image forming apparatus **1** from the front side is referred to as “right” or “right side”, and the left side when viewing the image forming apparatus **1** from the front side is referred to as “left” or “left side”. The terms “right side” and “left side” are collectively referred to as a left-right direction. The left-right direction in the present embodiment is a sheet width direction perpendicular to the sheet discharge direction D**1**.
(21) Further, in the following description and drawings, unless specifically stated otherwise, the term “connection” refers to the act of joining two members, regardless of a method used, such as fastening utilizing fasteners, welding of metal pieces, or adhesion utilizing adhesive substances. The term “fastening” refers to the use of fasteners such as a screw and a bolt to connect two separate members. To be noted, while, in the following description, it is described as two members to be fastened to each other, basically, other forms of connection techniques instead of the fastening can be used.

First Embodiment

(22) FIG. **1** is a schematic diagram illustrating a configuration of an image forming apparatus **1** of a first embodiment. A sheet processing apparatus **200** that processes a sheet S output from a printer body **100** is mounted on the top of the printer body **100**. Further, a document reading apparatus **300** that reads a document, and a document conveyance apparatus **400** that automatically conveys the document to a reading unit of the document reading apparatus **300** are mounted on the top of the sheet processing apparatus **200**.
(23) That is, the image forming apparatus **1** of the present embodiment is a multi-function machine including the printer body **100**, the sheet processing apparatus **200**, the document reading apparatus **300**, and the document conveyance apparatus **400**. That is, in addition to a function of forming an image on the sheet S in the printer body **100**, the image forming apparatus **1** possesses functions such as processing the sheet S with a formed image and reading the image from the document.
(24) To be noted, in the present embodiment, an image forming apparatus that includes the printer body **100** and at least one of the sheet processing apparatus **200**, the document reading apparatus **300**, and the document conveyance apparatus **400** is referred to as the “multi-function machine” or a “multi-function printer”. Further, an image forming apparatus that does not include any of the sheet processing apparatus **200**, the document reading apparatus **300**, and the document conveyance apparatus **400** is referred to as a “single-function printer”.
(25) As for the sheet S, serving as a recoding material (recording medium), it is possible to use various sheet materials which are different in size and materials, including paper such as standard paper and cardboard, a surface treated sheet material such as coated paper, a specially shaped sheet material such as an envelope and index paper, a plastic film, cloth, and the like.
(26) The printer body **100** (image forming apparatus body) includes an image forming portion **100**B and a casing **100**A accommodating the image forming portion **100**B. The sheet processing

apparatus **200** is an example of an upper unit arranged above the casing **100**A.
(27) The casing **100**A includes a body frame **100**F, described below. Further, the casing **100**A includes members (for example, a cover member forming an exterior surface of the printer body **100**, and a guide member forming a conveyance path of the sheet S in an interior of the printer body **100**) secured with respect to the body frame **100**F. Further, constituent elements, described below, of the printer body **100** may be built into the casing **100**A as a part of the casing, or may be made detachable with respect to the casing **100**A.
(28) Further, the discharge port **100***d* and the sheet discharge tray **135** are disposed in the casing **100**A. The discharge port **100***d* is an opening through which the sheet S, on which the image has been formed in the image forming portion **100**B, passes at the time when the sheet is discharged to the outside of the casing **100**A. A movement direction of the sheet S which is discharged from the discharge port **100***d* is referred to as the sheet discharge direction D**1**. The discharge port **100***d* is opened toward the front side of the image forming apparatus **1**.
(29) The sheet discharge tray **135** is a supporting portion by which the sheet S that has been discharged from the discharge port **100***d* is supported. The sheet discharge tray **135** is disposed on an upper surface portion of the casing **100**A. The sheet discharge tray **135** of the present embodiment is a surface which is disposed on the upper surface portion of the casing **100**A and is inclined upward toward the front side. In other words, the sheet discharge tray **135** is a surface that is inclined upward toward a downstream side in the sheet discharge direction D**1**.
(30) The image forming portion **100**B includes a photosensitive drum **102**, serving as an image bearing member, a charge roller **103**, serving as a charge unit, a laser scanner **104**, serving as an exposing unit, a developing roller **105**, serving as a developing unit, and a transfer roller **106**, serving as a transfer member. Further, the image forming portion **100**B includes a fixing unit **121**, serving as a fixing unit.
(31) The photosensitive drum **102** is a drum type electrophotographic photosensitive member. The photosensitive drum **102** is a member in which a photosensitive material such as an organic photo conductor (OPC), amorphous selenium, or amorphous silicon is formed on a circumferential surface of a cylindrical drum substrate made of such as aluminum and nickel. The photosensitive drum **102** is rotatably driven by a drive source arranged in the casing **100**A. The charge roller **103**, the developing roller **105**, and the transfer roller **106** are arranged around the photosensitive drum **102**. Further, the laser scanner **104** is arranged above the photosensitive drum **102**.
(32) A transfer nip **101**, serving as a transfer portion at which a toner image is transferred onto the sheet S, is formed between the transfer roller **106** and the photosensitive drum **102**. The transfer roller **106** is urged toward the photosensitive drum **102** by an urging member.
(33) The fixing unit **121** includes a fixing roller (heating roller), a pressing roller facing the fixing roller, and a heating unit such as a halogen lamp heating the fixing roller. The fixing unit **121** may be a film heating type in which a ceramic heater is arranged within an interior of a tubular shaped film.
(34) At least part of the image forming portion **100**B may be disposed in a cartridge that is detachable with respect to the casing **100**A. In the present embodiment, the photosensitive drum **102**, the charge roller **103**, and the developing roller **105** are disposed in a process cartridge **107** that is integrally detachable with respect to the casing **100**A.
(35) In the printer body **100**, a sheet feeding unit **111**, a conveyance roller pair **114**, a registration roller pair **115**, a post-fixing roller pair **132**, a sheet discharge roller pair **134**, the sheet discharge tray **135**, a reverse conveyance roller pair **136**, and a re-conveyance roller pair **137** are disposed. The sheet feeding unit **111** includes a cassette **112** in which the sheet S is stacked and stored, and a sheet feed unit **113** that feeds the sheet S from the cassette **112** one sheet at a time. Each of the sheet feeding unit **111**, the conveyance roller pair **114**, the registration roller pair **115**, the post-fixing roller pair **132**, the sheet discharge roller pair **134**, the reverse conveyance roller pair **136**, and the re-conveyance roller pair **137** can be said to include a roller that conveys the sheet S. In the

present embodiment, directions of rotation shafts of these rollers are aligned in parallel.
(36) The sheet discharge roller pair **134** is an example of a sheet discharge unit that discharges the sheet S, on which the image has been formed, to the outside of the casing **100**A of the printer body **100**. The sheet width direction is a direction parallel to rotational axes of rollers of the sheet discharge roller pair **134**. The sheet discharge roller pair **134** discharges the sheet S toward the front side of the image forming apparatus **1** via the discharge port **100***d* of the casing **100**A. The reverse conveyance roller pair **136** is an example of a reverse conveyance unit that reverses the sheet S at the time of duplex printing.
(37) The sheet processing apparatus **200** is an apparatus that performs processing on the sheet by receiving the sheet, on which the image has been formed by the image forming portion **100**B of the printer body **100**, from the casing **100**A of the printer body **100**. The sheet processing apparatus **200** is also called as a finisher.
(38) The sheet processing apparatus **200** includes a first sheet discharge roller pair **201**, a fixed sheet discharge tray **210**, an inlet roller pair **203**, a processing unit **204**, a second sheet discharge roller pair **207**, and a mobile sheet discharge tray **211**. Further, a branched conveyance path is disposed within a casing **200**A of the sheet processing apparatus **200**. The first sheet discharge roller pair **201** and the fixed sheet discharge tray **210** are arranged in a first conveyance path. The inlet roller pair **203**, the processing unit **204**, the second sheet discharge roller pair **211**, and the mobile sheet discharge tray **211** are arranged in a second conveyance path branched from the first conveyance path. The fixed sheet discharge tray **210** is secured with respect to the casing **200**A. The mobile sheet discharge tray is movable (liftable) with respect to the casing **200**A.
(39) The processing unit **204** includes a processing stage **206** supporting the sheet S and a sheet discharge roller **205** discharging the sheet S to the processing stage **206**. Further, the processing unit **204** includes a processing mechanism that is capable of performing at least one processing on the sheet S, including binding processing to bind a plurality of sheets of the sheet S or punching processing to create a hole in the sheet S.
(40) The document reading apparatus **300** is an image reading apparatus that reads image information from the document. The image forming apparatus **1** that includes the document reading apparatus **300** can form the image on the sheet based on the image information read by the document reading apparatus **300**.
(41) The document reading apparatus **300** includes a glass surface **301** on which the document is placed, and a reading unit **302** reading the image of the document. The document conveyance apparatus **400** includes a document tray **402** on which the document is stacked, a document feeding portion **401** feeding the document from the document tray **402**, a conveyance roller pair **403** conveying the fed document toward the reading unit **302**, and a conveyance guide **404** guiding the document.
(42) Operation of Image Forming Apparatus
(43) Next, an operation of the image forming apparatus **1** will be described. When an execution instruction (printing instruction) of an image forming operation is input to the image forming apparatus **1**, the image forming apparatus **1** starts an image forming operation. First, the photosensitive drum **102** is rotatably driven at a predetermined circumferential speed (process speed). The charge roller **103** uniformly charges a surface of the photosensitive drum **102** to a predetermined polarity and a predetermined potential. The laser scanner **104** exposes the photosensitive drum **102** by irradiating the surface of the photosensitive drum **102** with a laser light based on the image information input from an external computer along with the printing instruction, or the image information that the document reading apparatus **300** has read from the document. Thereby, a charge in exposed areas on the surface of the photosensitive drum **102** is removed, and an electrostatic latent image corresponding to the image information is formed. The developing roller **105** supplies toner, serving as developer, to the photosensitive drum **102**, and develops the electrostatic latent image to a toner image. The toner image formed on the

photosensitive drum **102** is conveyed toward the transfer nip **101** by the rotation of the photosensitive drum **102**.
(44) On the other hand, the sheet S is conveyed from the cassette **112** by the sheet feed unit **113** one sheet at a time. After passing through the conveyance roller pair **114** and receiving skew correction and a timing adjustment at the registration roller pair **115**, the fed sheet S is conveyed to the transfer nip **101**. Then, the toner image is transferred from the photosensitive drum **102** onto the sheet S at the transfer nip **101**.
(45) The sheet S passed through the transfer nip **101** is sent to the fixing unit **121**. While nipping and conveying the sheet S at a fixing nip between the fixing roller and the pressing roller, the fixing unit **121** fixes the toner image on the sheet S by applying heat and pressure to the toner image on the sheet S.
(46) The sheet S passed through the fixing unit **121** is guided by a first switching member **131** to a conveyance path that has been determined in the printing instruction in advance. In a case of performing simplex printing in which the image is formed on one side of the sheet S, the sheet S is guided by the first switching member **131** to the post-fixing roller pair **132**. In a case of performing duplex printing in which the image is formed on both surfaces of the sheet S, the sheet S on whose first surface the image has been formed is guided by the first switching member **131** to the reverse conveyance roller pair **136**. After having been reversed and conveyed (switchbacked) by the reverse conveyance roller pair **136**, the sheet S is again conveyed to the transfer nip **101** via the re-conveyance roller pair **137**. Then, the sheet S, on whose second surface opposite to the first surface the image has been formed by passing through the transfer nip **101** and the fixing nip, is guided by the first switching member **131** to the post-fixing roller pair **132**.
(47) In the present embodiment, the sheet S, which has completed either the simplex printing or the duplex printing, is conveyed via the post-fixing roller pair **132**. In a case where the sheet S is not sent to the sheet processing apparatus **200**, the sheet S that has passed through the post-fixing roller pair **132** is guided by a second switching member **133** to the sheet discharge roller pair **134**. In this case, the sheet S is discharged to the outside of the casing **100**A by the sheet discharge roller pair **134** by passing through the discharge port **100***d*, and is supported by the sheet discharge tray **135**. In a case where the sheet S is sent to the sheet processing apparatus **200**, the sheet S that has passed through the post-fixing roller pair **132** is guided by the second switching member **133** to a conveyance path heading to the sheet processing apparatus **200**. In this case, the sheet S is discharged to the outside of the casing **100**A through a discharge port communicating with the sheet processing apparatus **200**, and is received by the sheet processing apparatus **200**.
(48) The sheet S that the sheet processing apparatus **200** has received from the printer body **100** is guided by a switching member **202** to a conveyance path that has been determined in the printing instruction in advance. In a case where the processing is not performed on the sheet S, the sheet S is guided by the switching member **202** to the sheet discharge roller pair **201**, and is discharged to the fixed sheet discharge tray **210** by the sheet discharge roller pair **201**. In a case where the processing is to be performed on the sheet S, the sheet S is guided by the switching member **202** to the inlet roller pair **203**, and is conveyed toward the processing unit **204** by the inlet roller pair **203**. The processing unit **204** discharges the sheet S to the processing stage **206** by the sheet discharge roller **205**, and performs the processing such as the binding processing or the punching processing on the sheet S. The processed sheet S is discharged to the mobile sheet discharge tray **211** by the sheet discharge roller pair **207**.
(49) Operations of Document Reading Apparatus and Document Conveyance Apparatus
(50) Next, operations of the document reading apparatus **300** and the document conveyance apparatus **400** will be described. The document reading apparatus **300** and the document conveyance apparatus **400** can perform a feeding reading operation to read the image of the document while conveying the document by the document conveyance apparatus **400**. In the feeding reading operation, the document stacked on the document tray **402** is fed by a document

conveyance unit **401** one sheet at a time, and is guided by a conveyance guide **404**, so that the document is conveyed to the glass surface **301** of the document reading apparatus **300** via the conveyance roller pair **403**. The reading unit **302** of the document reading apparatus **300** optically scans the document through the glass surface **301**, and obtains the image information in which the image of the document is converted into electronic data. The document that has been read by the reading unit **302** is discharged to a document discharge tray disposed in the document conveyance apparatus **400**.

(51) Frame of Printer Body

(52) The body frame **100**F of the printer body **100** will be described. First, using FIGS. **2** and **3**, an overview of the body frame **100**F will be described. FIGS. **2** and **3** are perspective views illustrating part of frames of the image forming apparatus **1** of the first embodiment. To be noted, FIGS. **2** and **3** illustrate the body frame **100**F and an apparatus supporting member left **151** among the frames of the image forming apparatus **1**, which are illustrated in FIGS. **14** and **15**. As described below, the frames of the image forming apparatus **1** include the body frame **100**F, a frame **200**F of the sheet processing apparatus **200**, the apparatus supporting member left **151**, an apparatus supporting member right inner **260**, an apparatus supporting member right outer **261**, and the like.

(53) As illustrated in FIGS. **2** and **3**, the body frame **100**F includes a right side plate **140**, a left side plate **141**, a scanner stay **142**, a bottom stay **143**, a front stay **144**, a front upper stay **145**, a back upper stay **146**, and a back lower stay **147**. Further, the body frame **100**F includes an extension member right **148** connected to the right side plate **140**, and an extension member left **150** connected to the left side plate **141**. The right side plate **140** is an example of a first side plate. The left side plate **141** is an example of a second side plate. The front upper stay **145** is an example of a frame member. With respect to the rotational axis direction of the photosensitive drum **102**, the right side plate **140** is arranged on one end side of the process cartridge **107**, and the left side plate **141** is arranged on the other end side of the process cartridge **107**. The process cartridge **107** is attached and detached with respect to the casing **100**A in a direction intersecting with (preferably, perpendicularly intersecting with) the rotational axis direction of the photosensitive drum **102**. A guide member guiding the attachment of the process cartridge **107** with respect to the casing **100**A and a positioning portion positioning the process cartridge **107** at an image forming position are disposed in each of the right and left side plates **140** and **141**.

(54) The right side plate **140** is disposed on the right side of the body frame **100**F, and is a plate-shaped frame member spreading in the front-back direction and the up-down direction. That is, the right side plate **140** is arranged on a first side with respect to a center of the casing **100**A in the sheet width direction, and extends in a direction intersecting with the sheet width direction described above. The left side plate **141** is arranged on the left side of the body frame **100**F, and is a plate-shaped frame member spreading in the front-back direction and the up-down direction. That is, the left side plate **141** is arranged on a second side opposite to the first side with respect to the center of the casing **100**A in the sheet width direction, and extends in a direction intersecting with the sheet width direction described above. In other words, the right side plate **140** is arranged on the one end side with respect to the center of the casing **100**A, and the left side plate **141** is arranged on the other end side opposite to the one end side with respect to the center of the casing **100**A. Further, it is also possible to say that the right side plate **140** is arranged on the first side with respect to the center of the casing **100**A in the sheet width direction, and the left side plate **141** is arranged on the second side opposite to the first side with respect to the center of the casing **100**A in the sheet width direction. In other words, the right side plate **140** is arranged on the one end side with respect to the center of the apparatus body frame **100**F, and the left side plate **141** is arranged on the other end side opposite to the one end side with respect to the center of the apparatus body frame **100**F.

(55) The sheet feed unit **113**, the laser scanner **104**, and a core unit **160** of the printer body **100** are

arranged between the right and left side plates **140** and **141** in the left-right direction, and are each mounted to the apparatus body frame **100**F. The sheet feed unit **113** and the core unit **160** are fastened to the right and left side plates **140** and **141** with, for example, screws that penetrate the right and left side plates **140** and **141** from exterior to interior. That is, the right and left side plates **140** and **141** support the sheet feed unit **113** and the core unit **160**. The laser scanner **104** is placed on the scanner stay **142**, and is fastened to the scanner stay **142**. To be noted, the core unit **160** is a unit including the conveyance roller pair **114**, the registration roller pair **115**, the transfer roller **106**, and the re-conveyance roller pair **137** (refer to FIG. **1**). The core unit **160** forms conveyance paths through which the sheet S passes. In the present embodiment, the core unit **160** forms the conveyance paths for the sheet S during the execution of the simplex and duplex printing. Further, the core unit **160** includes a plurality of guide ribs that guide the sheet S passing through these conveyance paths.
(56) The scanner stay **142**, the bottom stay **143**, the front stay **144**, the front upper stay **145**, and the back lower stay **147** each are the frame member that connects the right and left side plates **140** and **141**. Details of the front upper stay **145**, and the back upper stay **146** will be described below.
(57) The scanner stay **142** is arranged above a mounting space into which the process cartridge **107** is mounted (also refer to FIG. **1**). The bottom stay **143** connects bottom portions of the right and left side plates **140** and **141** at positions further front than a central position of the casing **100**A in the front-back direction. The front stay **144** connects front portions of the right and left side plates **140** and **141**. The front upper stay **145** connects upper portions of the right and left side plates **140** and **141** at positions further front than the central position of the casing **100**A in the front-back direction. The back lower stay **147** connects back portions of the right and left side plates **140** and **141** at positions lower than a central position of the casing **100**A in the up-down direction.
(58) In the present embodiment, the scanner stay **142** and the bottom stay **143** are fastened to the right and left side plates **140** and **141** with screws that penetrate the right and left side plates **140** and **141** from exterior to interior. The front stay **144** is directly connected to a front surface of the left side plate **141**, and, concurrently, by being fastened to a front stay mounting plate **153** (refer to FIG. **2**) fastened to the right side plate **140**, is fastened to the right side plate **140** via the front stay mounting plate **153**. The back lower stay **147** is fastened to back surfaces of the right and left side plates **140** and **141**.
(59) Each stay (**142** to **147**) is a substantially planar-shaped metal member (metal plate) extending in the right-left direction. Each stay (**142** to **147**) may be subjected to drawing, hemming, or bending processing. The right and left side plates **140** and **141** are also substantially planar-shaped metal members (metal plates). The right and left side plates **140** and **141** may be subjected to the drawing, hemming, or bending processing.
(60) Further, as illustrated in FIG. **3**, an installation hole **141***a* for installing the fixing unit **121** is disposed in a back portion of the left side plate **141**. By disposing the installation hole **141***a*, when viewed from the back, a space between the right and left side plates **140** and **141** is expanded. Therefore, when assembling the printer body **100**, it becomes possible to mount the fixing unit **121** with respect to the body frame **100**F by inserting the fixing unit **121** from the back side of the body frame **100**F. A reinforcing plate **152** connecting upper and lower edges of the installation hole **141***a* in the back surface of the left side plate **141** is disposed on the left side plate **141**, and is fastened to the left side plate **141**. The left side plate **141** is reinforced by the reinforcing plate **152**, and a reduction in the strength of the body frame **100**F caused by the disposition of the installation hole **141***a* is mitigated.
(61) Extension Member
(62) Here, the extension member right **148** and the extension member left **150** will be described. By the inclusion or exclusion of the extension members, while sharing part of constituent elements with the image forming apparatus **1**, the image forming apparatus **1** of the present embodiment can be converted into other product forms that are different in function and performance.

(63) FIGS. **4** and **5** illustrate an image forming apparatus **2** as an example (variant example of the first embodiment) of the other product forms. The image forming apparatus **2** is a single function printer. FIG. **4** is a perspective view illustrating a frame **2**F of the image forming apparatus **2**. FIG. **5** is a schematic view illustrating the image forming apparatus **2**. Hereinafter, while comparing with the image forming apparatus **2** of the variant example, which does not include the extension member right **148** and the extension member left **150**, the extension member right **148** and the extension member left **150** in the image forming apparatus **1** of the present embodiment will be described.
(64) The image forming apparatus **2** of the variant example is different from the image forming apparatus **1** (refer to FIG. **1**) of the first embodiment in aspects such as a sheet conveyance path during the duplex printing and the number of sheets that can be supported by the sheet discharge tray **135**. Hereinafter, in the image forming apparatus **2**, members that are put with reference characters common to the image forming apparatus **1** of the first embodiment possess substantially the same configuration and function as corresponding members in the image forming apparatus **1** of the first embodiment.
(65) As illustrated in FIG. **4**, the frame **2**F of the image forming apparatus **2** of the variant example does not include the extension member right **148** and the extension member left **150** (refer to FIG. **2**). Further, the frame **2**F includes a back upper stay **1460** instead of the back upper stay **146** of the first embodiment, and includes a front upper stay **1450** instead of the front upper stay **145** of the first embodiment.
(66) In the image forming apparatus **2** of the variant example, the back upper stay **1460** is fastened to an upper surface and the back surface of the right side plate **140** and an upper surface and the back surface of the left side plate **141**.
(67) In contrast, in the image forming apparatus **1** of the first embodiment, as illustrated in FIGS. **2** and **3**, the extension member right **148** is mounted to a back portion of the upper surface of the right side plate **140**, and the extension member left **150** is mounted to a back portion of the upper surface of the left side plate **141**. The extension member right **148** is an example of a first extension member that is connected to the first side plate by being arranged above the first side plate. The extension member left **150** is an example of a second extension member that is connected to the second side plate by being arranged above the second side plate.
(68) The back upper stay **146** is fastened to an upper surface and a back surface of the extension member right **148**, and an upper surface and a back surface of the extension member left **150**. Within the upper surface of the right side plate **140**, a reinforcing plate **149** (refer to FIG. **2**) is disposed at a place to which the extension member right **148** is mounted. The reinforcing plate **149** suppresses the inclination of the extension member right **148** with respect to the right side plate **140** by reinforcing the upper surface of the right side plate **140**.
(69) As illustrated in FIG. **5**, the image forming apparatus **2**, which does not include the extension member, does not include the sheet discharge roller pair **134** and the reverse conveyance roller pair **136** (refer to FIG. **1**) of the first embodiment. The image forming apparatus **2** includes a triple roller **134**A, serving as a sheet discharge unit discharging the sheet S and a reverse unit reversing the sheet S at the time of the duplex printing. The triple roller **134**A includes a sheet discharge reverse roller **1340**, a sheet discharge idler roller **1341** coming into contact with the sheet discharge reverse roller **1340**, and a reverse idler roller **1342** coming into contact with the sheet discharge reverse roller **1340**.
(70) The sheet S passed through the fixing unit **121** is guided by a switching member **138** to a nip portion (sheet discharge nip) of the sheet discharge reverse roller **1340** and the sheet discharge idler roller **1341**, or a nip portion (reverse nip) of the sheet discharge reverse roller **1340** and the reverse idler roller **1342**. The sheet S that has completed either the simplex printing or the duplex printing is guided by the switching member **138** to the sheet discharge nip, and is discharged to the sheet discharge tray **135** by the sheet discharge reverse roller **1340** and the sheet discharge idler roller

1341. In a case of performing the duplex printing, the sheet S on whose first surface the image has been formed is guided by the switching member **138** to the reverse nip. After having been reversed and conveyed (switchbacked) by the sheet discharge roller **1340** and the reverse idler roller **1342**, the sheet S is again guided to the transfer nip **101** via the re-conveyance roller pair **137**. Then, the sheet S on whose second surface opposite to the first surface the image has been formed by passing through the transfer nip **101** and the fixing nip is guided by the switching member **138** to the sheet discharge nip.
(71) When comparing FIGS. **2** and **4**, in comparison with the frame **2**F that does not include the extension members, in the body frame **100**F of the first embodiment that includes the extension members, a back portion of the body frame **100**F is extended upward. Therefore, in comparison with the sheet discharge reverse roller **1340** and the sheet discharge idler roller **1341** (refer to FIG. **5**) of the image forming apparatus **2** that does not include the extension members, it is possible to arrange the sheet discharge roller pair **134** (refer to FIG. **1**) of the image forming apparatus **1** of the first embodiment at a higher position. That is, at least part of the sheet discharge roller pair **134** is arranged above the upper surfaces of the right and left side plates **140** and **141** by utilizing a space of the casing **100**A that is expanded by the extension member right **148** and the extension member left **150**. Thereby, it is possible to increase the number of sheets that can be supported by the sheet discharge tray **135**.
(72) To be noted, when the position of the sheet discharge roller pair **134** is elevated, depending on the velocity of the sheet S at the time of discharge, there is a possibility that the sheet S falls by jumping over the sheet discharge tray **135**. Therefore, as described below, by increasing an inclination angle α (refer to FIGS. **1** and **5**), the sheet S can be more reliably landed onto the sheet discharge tray **135**. The inclination angle α is an angle of an upper surface of the sheet discharge tray **135** with respect to a horizontal surface in a case where the image forming apparatus **1** or **2** is viewed in the sheet width direction (left-right direction).
(73) Further, in comparison with a conveyance distance from the fixing unit **121** to the sheet discharge nip in the image forming apparatus **2** (refer to FIG. **5**) that does not include the extension members, a conveyance distance from the fixing unit **121** to the sheet discharge roller pair **134** in the image forming apparatus **1** (refer to FIG. **1**) that includes the extension members is longer. In the image forming apparatus **1** (refer to FIG. **1**) of the first embodiment, in addition to the sheet discharge roller pair **134**, it is possible to arrange the reverse conveyance roller pair **136** dedicated for reversing the sheet S during the duplex printing. Therefore, by drive controlling the timing and a drive speed of the reverse conveyance roller pair **136** independently from the sheet discharge roller pair **134**, it is possible to improve productivity in comparison with the image forming apparatus **2** that does not include the extension members.
(74) Front Upper Stay
(75) Next, while comparing with the front upper stay **1450** of the image forming apparatus **2**, the front upper stay **145** in the image forming apparatus **1** of the first embodiment will be described.
(76) As illustrated in FIG. **4**, in the image forming apparatus **2**, the front upper stay **1450** is fastened to the upper surfaces of the right and left side plates **140** and **141**. The front upper stay **1450** is arranged downstream of the sheet discharge tray **135** in the sheet discharge direction D**1**. Here, the term “downstream of the sheet discharge tray **135** in the sheet discharge direction D**1**” refers to a position where a distance to a downstream end of the sheet discharge tray **135** in the sheet discharge direction D**1** is shorter than a distance to an upstream end of sheet discharge tray **135** in the sheet discharge direction D**1**. That is, the front upper stay **1450** is arranged at the position where the distance to the downstream end of the sheet discharge tray **135** in the sheet discharge direction D**1** is shorter than the distance to the upstream end of sheet discharge tray **135** in the sheet discharge direction D**1**. Further, the sheet discharge tray **135** is a surface that is located on an upper surface of the printer body **100**, and is inclined upward toward the front side.
(77) The front upper stay **1450** is a substantially planar-shaped member extending in the left-right

direction. The front upper stay **1450** may be subjected to the drawing and hemming processing for reinforcement.

(78) As illustrated in FIG. **2**, in the image forming apparatus **1** of the first embodiment, the front upper stay **145** is fastened to the upper surfaces of the right and left side plates **140** and **141**. FIG. **6** is a schematic diagram illustrating a positional relationship between the sheet discharge tray **135** and the front upper stay **145**, and is a cross-sectional view of the casing **100**A taken along a plane perpendicular to the left-right direction.

(79) As illustrated in FIG. **6**, the front upper stay **145** is arranged downstream of the discharge port **100***d* of the casing **100**A in the sheet discharge direction D**1**. The front upper stay **145** is arranged downstream of the sheet discharge tray **135** in the sheet discharge direction D**1**. That is, the front upper stay **145** is arranged at the position where the distance to the downstream end of the sheet discharge tray **135** in the sheet discharge direction D**1** is shorter than the distance to the upstream end of sheet discharge tray **135** in the sheet discharge direction D**1**. The sheet discharge tray **135** is inclined upward toward the downstream side in the sheet discharge direction D**1**, and is extended to a position higher than an upper end **140***t* of the right side plate **140** and an upper end **141***t* of the left side plate **141**. That is, the downstream end of the sheet discharge tray **135** is positioned above the front upper stay **145**. Further, among end portions of the casing **100**A, an end portion that is positioned upstream of the discharge port **100***d* in the sheet discharge direction D**1** is referred to as an upstream end, and an end portion that is positioned downstream of the discharge port **100***d* in the sheet discharge direction D**1** is referred to as a downstream end. In this case, as illustrated in FIG. **6**, when viewed in the sheet width direction, a position of the front upper stay **145** is closer to the downstream end of the casing **100**A than the upstream end of the casing **100**A. That is, a distance between the front upper stay **145** and the downstream end of the casing **100**A is shorter than a distance between the front upper stay **145** and the upstream end of the casing **100**A.

(80) The front upper stay **145** includes two side plate connection surfaces **145***d*, and supporting member connection surfaces **145***a* and **145***b*. The front upper stay **145** is connected to the right and left side plates **140** and **141** via each of the side plate connection surfaces **145***d*, and is connected to the apparatus supporting member right inner **260** (refer to FIG. **14**), described below, and the apparatus supporting member left **151** via the supporting member connection surfaces **145***a* and **145***b*, respectively. Further, the front upper stay **145** includes an intermediate portion **145***c* that connects the side plate connection surface **145***d* on the right and the supporting member connection surface **145***b* with the side plate connection surface **145***d* on the left and the supporting member connection surface **145***a*.

(81) The side plate connection surface **145***d* connected to the right side plate **140** is a first surface of the front upper stay **145**. The side plate connection surface **145***d* connected to the left side plate **141** is a second surface of the front upper stay **145**. The supporting member connection surface **145***b* connected to the apparatus supporting member right inner **260** is a third surface of the front upper stay **145**. The supporting member connection surface **145***a* connected to the apparatus supporting member left **151** is a fourth surface of the front upper stay **145**.

(82) The side plate connection surface **145***d* on the right is connected to the upper surface of the right side plate **140**. The side plate connection surface **145***d* on the right and the upper surface of the right side plate **140** are surfaces that spread in a direction intersecting with the vertical direction, and, in the present embodiment, are surfaces that spread in the horizontal direction. The side plate connection surface **145***d* on the left is connected to the upper surface of the left side plate **141**. The side plate connection surface **145***d* on the left and the upper surface of the left side plate **141** are surfaces that spread in a direction intersecting with the vertical direction, and, in the present embodiment, are surfaces that spread in the horizontal direction. The supporting member connection surface **145***b* on the right is connected to the apparatus supporting member right inner **260** at a position above the upper surface of the right side plate **140**. The supporting member connection surface **145***a* on the left is connected to the apparatus supporting member left **151** at a

position above the upper surface of the left side plate **141**.

(83) The side plate connection surfaces **145***d* and the supporting member connection surfaces **145***a* and **145***b* of the front upper stay are formed by a single metal plate. The metal plate is bent between the side plate connection surface **145***d* on the right extending substantially in the front-back direction and the supporting member connection surface **145***b* on the right extending substantially in the up-down direction, and is bent between the side plate connection surface **145***d* on the left extending substantially in the front-back direction and the supporting member connection surface **145***a* on the left extending substantially in the up-down direction.

(84) As illustrated in FIG. **16**, the intermediate portion **145***c* includes a surface **145***e* (fifth surface) that is continuous to the side plate connection surfaces **145***d* on the left and right, and a surface **145***f* (sixth surface) that is continuous to the supporting member connection surfaces **145***a* and **145***b* on the left and the right. The surface **145***e* is a surface that spreads in a direction intersecting with the vertical direction. In the present embodiment, the surface **145***e* is a surface that extends in the left-right direction so as to connect the side plate connection surfaces **145***d* on the left and right, and spreads substantially in the front-back direction (horizontal direction). The surface **145***f* is a surface that spreads in a direction intersecting with the horizontal direction. In the present embodiment, the surface **145***f* is a surface that extends in the left-right direction so as to connect the supporting member connection surfaces **145***a* and **145***b* on the left and right, and spreads substantially in the up-down direction (vertical direction).

(85) The surfaces **145***e* and **145***f* of the intermediate portion **145***c* are formed by the same metal plate as the side plate connection surfaces **145***d* and the supporting member connection surfaces **145***a* and **145***b*. When viewed in the left-right direction, the metal plate is bent between the surface **145***e* extending substantially in the front-back direction and the surface **145***f* extending substantially in the up-down direction. By shaping a cross-sectional shape of the intermediate portion **145***c* into a bent shape when viewed in the left-right direction, in comparison with a case where the cross-sectional shape of the intermediate portion **145***c* is linear, an effect of increasing the rigidity of the body frame **100**F through the use of the front upper stay **145** can be further enhanced by reinforcing the rigidity of the front upper stay **145** itself. Further, since the surface **145***f* is a surface that is continuous to the supporting member connection surfaces **145***a* and **145***b*, it is possible to more stably support the upper unit (sheet processing apparatus **200**) that is supported via the supporting member connection surfaces **145***a* and **145***b*.

(86) To be noted, as illustrated in FIG. **16**, the intermediate portion **145***c* can be formed into a rectangular one side open shaped cross-section in which the surface **145***e* extends upward from a front end of the surface **145***e* and a surface **145***g* extends upward from a back end of the surface **145***e*. Thereby, it is possible to further increase the rigidity of the front upper stay **145** and the body frame **100**F.

(87) As illustrated in FIG. **6**, when viewed in the left-right direction, an upper end **145***cl* of the intermediate portion **145***c* of the front upper stay **145** is positioned at a position higher than an upper end **140***t* of the right side plate **140** and an upper end **141***t* of the left side plate **141**. Further, the intermediate portion **145***c* of the front upper stay **145** is positioned below the sheet discharge tray **135**. That is, in the present embodiment, when viewed in the sheet width direction, the upper end of the intermediate portion of the frame member is positioned above the upper surfaces of the first and second side plates and below the supporting portion.

(88) Further, the intermediate portion **145***c* of the front upper stay **145** is arranged between the right and left plates **140** and **141** in the left-right direction. The intermediate portion **145***c* of the front upper stay **145** may support a lower surface of the sheet discharge tray **135**.

(89) The supporting member connection surfaces **145***a* and **145***b* are arranged outside of the sheet discharge tray **135** in the left-right direction. That is, in the left-right direction, the sheet discharge tray **135** is arranged between the supporting member connection surfaces **145***a* and **145***b*. At least one of the upper ends of the supporting member connection surfaces **145***a* and **145***b* projects above

the upper end **145**_cl_ of the surface **145**_f_ of the intermediate portion **145**_c_ (refer to FIG. **2**). That is, at least one of the upper ends of the third surface and the fourth surface of the frame member is positioned above the upper end of the intermediate portion of the frame member. In the present embodiment, both the upper ends of the supporting member connection surfaces **145**_a_ and **145**_b_ project above the upper end **145**_cl_ of the surface **145**_f_ of the intermediate portion **145**_c_.

(90) To be noted, it is preferable to enhance the rigidity of the front upper stay **145** by increasing a plate thickness in comparison with the front upper stay **1450** illustrated in FIG. **4**. It is preferable to enhance the rigidity of the front upper stay **145** by increasing the plate thickness in comparison with the right and left side plates **140** and **141**. The front upper stay **145** may be subjected to the drawing and the hemming processing for reinforcement.

(91) Apparatus Supporting Member Left

(92) Next, the apparatus supporting member left **151** that is arranged on the upper surface of the left side plate **141** in the image forming apparatus **1** of the first embodiment will be described. As illustrated in FIG. **2**, the apparatus supporting member left **151** is arranged between the extension member left **150** and the front upper stay **145** in the front-back direction. The apparatus supporting member left **151** is connected to the upper surfaces of the extension member left **150** and the left side plate **141**, and is connected to the supporting member connection surface **145**_a_ of the front upper stay **145**.

(93) As illustrated in FIG. **13**, positioning projections **151**_a_ and **151**_b_ are disposed on the apparatus supporting member left **151**. The positioning projections **151**_a_ and **151**_b_ are in a projecting shape projecting forward from a front surface of the apparatus supporting member left **151**, and, for example, is formed by drawing a part of a metal plate that forms the apparatus supporting member left **151**.

(94) Positioning holes **145**_a_**1** and **145**_a_**2** that respectively engage with the positioning projections **151**_a_ and **151**_b_ of the apparatus supporting member left **151** are disposed in the supporting member connection surface **145**_a_ of the front upper stay **145**. By engaging the positioning holes **145**_al_ and **145**_a_**2** with the positioning projections **151**_a_ and **151**_b_, the inclination of the extension member left **150** with respect to the upper surface of the left side plate **141** is suppressed.

(95) Image Processing Unit

(96) Next, an image processing unit **505** that is included in the image forming apparatus **1** of the first embodiment will be described. The image forming apparatus **1** of the present embodiment is configured to be capable of mounting the document reading apparatus **300** and the document conveyance apparatus **400** on the upper part of the printer body **100** by replacing an image processing unit from the image forming apparatus **2** (refer to FIGS. **4** and **5**) of the variant example and also by changing a position of a fan.

(97) FIG. **7**A is a schematic diagram illustrating a state in which the frame **2**F of the variant example is viewed from the left. FIG. **7**B is a schematic diagram illustrating a state in which the body frame **100**F of the first embodiment is viewed from the left.

(98) As illustrated in FIG. **7**A, the image forming apparatus **2** includes fans **502** and **503**, an operation control unit **500**, and an image processing unit **501**. The fans **502** and **503** draw in exterior air, and circulate the air to an interior of the apparatus. The operation control unit **500** is a circuit substrate including a control circuit for controlling an operation (for example, image forming operation described above) of the image forming apparatus **2**. The image processing unit **501** is a circuit substrate including a control circuit. This control circuit, for example, processes the image information sent from an external computer along with the printing instruction, and converts the image information into data for driving the laser scanner **104**.

(99) In the image forming apparatus **2**, the fans **502** and **503**, the operation control unit **500**, and the image processing unit **501** are arranged on an outer side (left side) of the left side plate **141**. The fan **502** is arranged above the operation control unit **500**, and the fan **503** is arranged above the image processing unit **501**. The fans **502** and **503**, the operation control unit **500**, and the image

processing unit **501** are all positioned within a silhouette (rectangular outline) of the left side plate **141** when viewed from the left.

(100) On the other hand, in the image forming apparatus **1** of the first embodiment that is a multi-function machine, so as to process the image information read by the document reading apparatus **300** and so as to accomplish the image processing for facsimile functionality, in comparison with the image forming apparatus **2** that is a single-function printer, a processing capacity required for the image processing unit is greater. Therefore, the image forming apparatus **1** of the first embodiment includes the image processing unit **505** that is larger than the image processing unit **501** (refer to FIG. **7**A) of the variant example.

(101) As illustrated in FIG. **7**B, when viewed from the left, the image processing unit **505** of the first embodiment does not fit within the silhouette (rectangular outline) of the left side plate **141**. In particular, part of an upper portion of the image processing unit **505** projects above the silhouette of the left side plate **141**.

(102) In the image forming apparatus **1** of the first embodiment, the extension member left **150** is disposed on the upper surface of the left side plate **141**. The width of the image processing unit **505** in the up-down direction is contained within a range from a lower end of the left side plate **141** to an upper end of the extension member left **150**.

(103) Further, the image forming apparatus **1** of the first embodiment includes a fan **504** instead of the fan **503** (refer to FIG. **7**A) of the image forming apparatus **2** of the variant example. When viewed from the left, the fan **504** is arranged inside of the apparatus supporting member left **151**. To be noted, the apparatus supporting member left **151** includes a substantially rectangularly shaped recess portion when viewed from the left. This recess portion is opened toward the left, and houses the fan **504**. The image processing unit **505** is arranged at the back of the operation control unit **500**. At least part of the apparatus supporting member left **151** is positioned at the back of a back end of the operation control unit **500** in the front-back direction. Further, at least part of the apparatus supporting member left **151** is positioned in front of a front end of the image processing unit **505** in the front-back direction.

(104) Next, using FIGS. **8** and **9**, an image processing unit case **170**, serving as a case member to which the image processing unit **505** is attached, will be described. FIG. **8** is a perspective view illustrating the body frame **100**F to which the image processing unit case **170** is mounted. FIG. **9** is a perspective view illustrating the body frame **100**F before the image processing unit case **170** is mounted. The image processing unit case **170** is a part of the casing **100**A of the printer body **100**.

(105) As illustrated in FIGS. **8** and **9**, a case body **170***a* and an interface plate **171** are integrated in the image processing unit case **170**, and the image processing unit case **170** is formed substantially in a box shape. A plurality of holes are formed in the interface plate **171** to accommodate the attachment of connectors, such as a local area network (LAN) connector and a universal serial bus (USB) connector disposed in the image processing unit **505**, with respect to an interface unit.

(106) Part of the image processing unit case **170** projects above the upper surface of the left side plate **141**. A mounting plate **162** is disposed to the left side plate **141**. By fastening the case body **170***a* to the mounting plate **162** and by fastening the interface plate **171** to the back surface of the left side plate **141**, the image processing unit case **170** is secured to the left side plate **141**. Further, the image processing unit case **170** is fastened with respect to each of the extension member left **150** and the apparatus supporting member left **151**.

(107) As described above, in the present embodiment, the apparatus supporting member left **151** and the left side plate **141** are connected via the image processing unit case **170**. In other words, the case member is connected to each of the second side plate and the second supporting member. Therefore, in comparison with a configuration in which the apparatus supporting member left **151** and the left side plate **141** are not connected via the image processing unit case **170**, the inclination of the apparatus supporting member left **151** with respect to the left side plate **141** is suppressed. By suppressing the inclination of the apparatus supporting member left **151**, the inclination of the

upper unit (sheet processing apparatus **200**) that is supported by the apparatus supporting member left **151** is also suppressed.
(108) Further, since the image processing unit case **170** is further connected also to the extension member left **150**, the extension member left **150** and the left side plate **141** are connected via the image processing unit case **170**. Therefore, in comparison with a configuration in which the extension member left **150** and the left side plate **141** are not connected via the image processing unit case **170**, the inclination of the extension member left **150** with respect to the left side plate **141** is suppressed. By suppressing the inclination of the extension member left **150**, the inclination of the upper unit (sheet processing apparatus **200**) that is supported by the extension member left **150** is also suppressed.
(109) Frame Configuration of Sheet Processing Apparatus
(110) Using FIGS. **10** and **11**, a frame configuration of the sheet processing apparatus **200** will be described. FIGS. **10** and **11** are perspective views illustrating the frame **200**F of the sheet processing apparatus **200**.
(111) The frame **200**F of the sheet processing apparatus **200** includes a right side plate **220**, a left side plate **221**, a top plate **224**, a bottom plate **222**, and a front bottom stay **223**. The right, left, top, and bottom plates **220**, **221**, **224**, and **222** and the front bottom stay **223** are fastened to one another, and form a structure substantially identical to a box shape.
(112) Further, the frame **200**F includes an electrical component installation plate **225**, a front plate **226**, and a reinforcing plate **227**. The electrical component installation plate **225** and the front plate **226** are mounted to a front surface of the left side plate **221**, and are arranged at positions projecting forward with respect to the box shape formed by the right, left, top, bottom plates **220**, **221**, **224**, and **222** and the front bottom stay **223**. The reinforcing plate **227** is fastened to upper surfaces of the light side plate **221** and the electrical component installation plate **225**. The left side plate **221** and the electrical component installation plate **225** are firmly connected through the reinforcing plate **227**.
(113) A right side mounting portion **230** and a left side mounting portion **240** are disposed to a lower part of the sheet processing apparatus **200**. The right side and left side mounting portions **230** and **240** are mounting portions for mounting the sheet processing apparatus **200** onto the printer body **100**. The right side and left side mounting portions **230** and **240** are positioning portions for positioning the sheet processing apparatus **200** with respect to the printer body **100**.
(114) The right side mounting portion **230** includes a right side mounting member **231** made from resin. On the right side mounting member **231**, a first positioning shaft **232** that engages with the printer body **100** projects. The left side mounting portion **240** includes a left side mounting member **241**, which is formed into a box shape using a metal plate. On the left side mounting member **241**, a second positioning shaft **242** that engages with the printer body **100** projects.
(115) Connection Configuration of Sheet Processing Apparatus and Document Reading Apparatus
(116) Using FIG. **12**, a connection configuration of the sheet processing apparatus **200** and the document reading apparatus **300** will be described. FIG. **12** is a schematic diagram illustrating the sheet processing apparatus **200** and the document reading apparatus **300** as viewed from the left.
(117) As illustrated in FIG. **12**, a base **310** is disposed on the top plate **224** of the sheet processing apparatus **200**. The base **310** may be pre-assembled into the sheet processing apparatus **200**, or may be a separate member from both the sheet processing apparatus **200** and the document reading apparatus **300**. The base **310** is a supporting member that supports a unit disposed above the sheet processing apparatus **200**. The base **310** is made from resin, and also incorporates functions such as a cable guide to guide cables that connect the document reading apparatus **300** and the sheet processing apparatus **200** or the printer body **100**. In the present embodiment, the document reading apparatus **300** is installed on the base **310**, and the document reading apparatus **300** and the sheet processing apparatus **200** are integrated via the base **310**.
(118) To be noted, the document conveyance apparatus **400** is integrated with the document reading

apparatus **300** in advance by being connected to the document reading apparatus **300** via, for example, a hinge portion. That is, by mounting the document reading apparatus **300** onto the base **310**, also the document conveyance apparatus **400** is mounted to the image forming apparatus **1**.

(119) Connection Configuration of Printer Body and Sheet Processing Apparatus

(120) Using FIGS. **13** to **16**, a connection configuration of the printer body **100** and the sheet processing apparatus **200** will be described. FIG. **13** is a perspective view illustrating the extension member right **148**, the extension member left **150**, and the apparatus supporting member left **151**, and corresponds to an enlarged view of an upper part of FIG. **2**. FIGS. **14**, **15**, and **16** are perspective views illustrating a frame configuration of the image forming apparatus **1** with the sheet processing apparatus **200** installed on the top of the printer body **100**.

(121) First, a positioning configuration of the sheet processing apparatus **200** and the printer body **100** will be described. As illustrated in FIG. **13**, a positioning hole right **252** and a positioning hole left **253**, serving as positioning portions for positioning the sheet processing apparatus **200**, are disposed in the printer body **100**. The positioning hole right **252** is formed in the upper surface of the extension member right **148**, and the positioning hole left **253** is formed in the upper surface of the extension member left **150**.

(122) The sheet processing apparatus **200** is configured such that, by respectively engaging the first and second positioning shafts **232** (refer to FIG. **11**) and **242** with the positioning hole right **252** and the positioning hole left **253**, a position of the sheet processing apparatus **200** is determined with respect to the printer body **100**.

(123) Next, the connection of the sheet processing apparatus **200** and the printer body **100** will be described. As illustrated in FIG. **13**, a plurality of brackets **251***a* to **251***e* for fastening the sheet processing apparatus **200** and the printer body **100** are installed on the extension member right **148**, the extension member left **150**, and the apparatus supporting member left **151**. In particular, the brackets **251***a* and **251***b* are fastened to the upper surface of the extension member right **148**. The bracket **251***c* is fastened to the upper surface of the extension member left **150**. The brackets **251***d* and **251***e* are fastened to the upper surface of the apparatus supporting member left **151**. Further, a fastening portion **254** for fastening the sheet processing apparatus **200** and the printer body **100** is disposed also on an upper end portion of the image processing unit case **170**.

(124) The sheet processing apparatus **200** is installed on the extension member right **148**, the extension member left **150**, and the apparatus supporting member left **151**. The right side mounting portion **230** (refer to FIGS. **10** and **11**) of the sheet processing apparatus **200** is fastened to the brackets **251***a* and **251***b*. The left side mounting portion **240** (refer to FIGS. **10** and **11**) of the sheet processing apparatus **200** is fastened to the bracket **251***c* and the fastening portion **254** of the image processing unit case **170**. The front plate **226** (refer to FIGS. **10** and **11**) of the sheet processing apparatus **200** is fastened to the brackets **251***d* and **251***e*. As described above, the sheet processing apparatus **200** is connected to the printer body **100** via the brackets **251***a* to **251***e* and the image processing unit case **170**.

(125) As illustrated in FIG. **14**, the apparatus supporting member right inner **260** and an apparatus supporting member right outer **261** are disposed on the upper surface of the right side plate **140** between the extension member right **148** and the front upper stay **145**.

(126) The apparatus supporting member right inner **260** is an example of a first supporting member, and the apparatus supporting member left **151** is an example of a second supporting member. The apparatus supporting member right inner **260** is formed by one or more metal plates. The apparatus supporting member right inner **260** may be formed by a bent metal plate, and may be formed by a plurality of metal plates. The apparatus supporting member left **151** is formed by one or more metal plates. The apparatus supporting member left **151** may be formed by a bent metal plate, and may be formed by a plurality of metal plates. The apparatus supporting member right inner **260** supports the sheet processing apparatus **200** on the right side of the image forming apparatus **1**. That is, the apparatus supporting member right inner **260** supports the upper unit on the first side in the sheet

width direction. The apparatus supporting member left **151** supports the sheet processing apparatus **200** on the left side of the image forming apparatus **1**. That is, the apparatus supporting member left **151** supports the upper unit on the second side opposite to the first side in the sheet width direction.
(127) To be noted, the first and second supporting members are not limited to a member that is directly connected to the upper unit, but may be a member that supports the upper unit by being connected to the other member which is connected to the upper unit. In the present embodiment, the apparatus supporting member right inner **260**, serving as the first supporting member, supports the sheet processing apparatus **200** by being connected to the apparatus supporting member right outer **261** connected to the sheet processing apparatus **200**. The apparatus supporting member right inner **260** and the apparatus supporting member right outer **261** may be formed as an integrated member.
(128) The apparatus supporting member right inner **260** is mounted to the upper surface of the right side plate **140**, and is also mounted to the supporting member connection surface **145***b* of the front upper stay **145**. More particularly, the apparatus supporting member right inner **260** is fastened to the upper surface of the right side plate **140**, and is fastened to supporting member connection surface **145***b* of the front upper stay **145**. Further, the apparatus supporting member right inner **260** is fastened to the bracket **251***b* (refer to FIG. **16**).
(129) The apparatus supporting member right outer **261** is fastened to a right surface of the apparatus supporting member right inner **260**. Further, an upper connection surface **261***a* of the apparatus supporting member right outer **261** is connected to the frame **200**F of the sheet processing apparatus **200**. Further, the apparatus supporting member right outer **261** is also fastened to the right side mounting member **231** (refer to FIGS. **10** and **11**) of the sheet processing apparatus **200**.
(130) As described above, in the present embodiment, the frame **200**F of the sheet processing apparatus **200** and the body frame **100**F of the printer body **100** are connected via the apparatus supporting member right inner **260** and the apparatus supporting member right outer **261**. Therefore, in comparison with a case where the frame **200**F of the sheet processing apparatus **200** and the body frame **100**F of the printer body **100** are not connected via the apparatus supporting member right inner **260** and the apparatus supporting member right outer **261**, it is possible to more firmly connect the sheet processing apparatus **200** and the printer body **100**.
(131) Further, the upper connection surface **261***a* of the apparatus supporting member right outer **261** is fastened to the top plate **224** of the sheet processing apparatus **200**. That is, the apparatus supporting member right inner **260** is connected to the upper surface of the sheet processing apparatus **200** via the apparatus supporting member right outer **261** that is connected to the apparatus supporting member right inner **260**. With this configuration, it is possible to more firmly connect the sheet processing apparatus **200** and the printer body **100**. To be noted, it is acceptable if at least one of the first and second supporting members is connected to the upper surface of the upper unit.
(132) Further, as illustrated in FIG. **14**, the image forming apparatus **1** includes a connecting plate **250**. The connecting plate **250** is fastened to each of the plate **226** of the sheet processing apparatus **200** and a front surface of the apparatus supporting member left **151**, and connects the front plate **226** and the apparatus supporting member left **151**.
(133) As described above, in the present embodiment, the frame **200**F of the sheet processing apparatus **200** and the body frame **100**F of the printer body **100** are connected via the apparatus supporting member left **151** and the connecting plate **250**. Therefore, in comparison with a case where the frame **200**F of the sheet processing apparatus **200** and the body frame **100**F of the printer body **100** are not connected via the apparatus supporting member left **151** and the connecting plate **250**, it is possible to more firmly connect the sheet processing apparatus **200** and the printer body **100**.
(134) As described above, the sheet processing apparatus **200** is installed on the extension member

right **148**, the extension member left **150**, and the apparatus supporting member left **151**, and is connected to the printer body **100** via the plurality of brackets **251***a* to **251***e* and the image processing unit case **170**. On the back side of the image forming apparatus **1**, a part (i.e., upstream part in the sheet discharge direction D**1**) of the sheet processing apparatus **200** is supported by the extension member right **148** and the extension member left **150**, and, on the front side of the image forming apparatus **1**, another part (i.e., downstream part in the sheet discharge direction D**1**) of the sheet processing apparatus **200** is supported by the apparatus supporting member right inner **260** and the apparatus supporting member left **151**.
(135) The apparatus supporting member right inner **260**, serving as the first supporting member, is disposed on a right front side of the image forming apparatus **1**. The apparatus supporting member right inner **260** is installed on the upper surface of the right side plate **140** of the body frame **100**F, and is fastened to the supporting member connection surface **145***b* that is an upright surface of the front upper stay **145**. Thereby, the inclination of the apparatus supporting member right inner **260** with respect to the right side plate **140** is suppressed. Then, the inclination of the sheet processing apparatus **200** itself with respect to the right side plate **140** is also suppressed by fastening the apparatus supporting member right outer **261** to the apparatus supporting member right inner **260** and the top plate **224** of the sheet processing apparatus **200**.
(136) The apparatus supporting member left **151**, serving as the second supporting member, is disposed on a left front side of the image forming apparatus **1**. The apparatus supporting member left **151** is mounted to the upper surface of the left side plate **141** of the body frame **100**F and the upper surface of the extension member left **150**, and is mounted to the supporting member connection surface **145***a* that is the upright surface of the front upper stay **145**. More particularly, the apparatus supporting member left **151** is installed on the upper surface of the left side plate **141** of the body frame **100**F and the upper surface of the extension member left **150**, and is fastened to the supporting member connection surface **145***a* that is the upright surface of the front upper stay **145**. By engaging the positioning projections **151***a* and **151***b* of the apparatus supporting member left **151** with the positioning holes **145***a***1** and **145***a***2** of the supporting member connection surface **145***a*, the inclination of the apparatus supporting member left **151** with respect to the left side plate **141** is suppressed. Then, the sheet processing apparatus **200** is connected to the apparatus supporting member left **151** via the brackets **251***d* and **251***e*, and, on the front side of the apparatus, the apparatus supporting member left **151** and the front plate **226** of the sheet processing apparatus **200** are connected by the connecting plate **250**. Thereby, the inclination of the sheet processing apparatus **200** itself with respect to the left side plate **141** is also suppressed.
(137) According to the present embodiment described above, the supporting member connection surfaces **145***a* and **145***b* of the front upper stay **145** that is arranged on the front side of the image forming apparatus **1** are respectively connected to the apparatus supporting member left **151** and the apparatus supporting member right inner **260**. That is, the front upper stay **145** (frame member) that is arranged downstream of the discharge port **100***d* in the sheet discharge direction D**1** includes the third surface (supporting member connection surface **145***b*) that is connected to the apparatus supporting member right inner **260**, and the fourth surface (supporting member connection surface **145***a*) that is connected to the apparatus supporting member left **151**. Therefore, it becomes possible to increase the rigidity of the image forming apparatus in its downstream part in the sheet discharge direction.
(138) A user normally stands in front of the image forming apparatus **1** to perform operations such as opening and closing the document conveyance apparatus **400** and removing the sheet from the sheet discharge tray **135**. At that time, if the casing **100**A of the printer body **100** and the upper unit do not seem integrated, it may give an impression of insufficient rigidity.
(139) Further, the image forming apparatus **1** is equipped with wheels at the bottom of the apparatus for transporting the whole equipment, or the whole equipment is mounted on a cart. When moving the whole equipment, the user or an operator may sometimes place hands on the

document reading apparatus **300** or the sheet processing apparatus **200** to overcome steps or gaps, or to move across inclined surfaces. At this time, if the frame of the sheet processing apparatus **200** becomes twisted or tilted, there is a possibility that an impression of a lack of unity and rigidity between the casing **100**A of the printer body **100** and the upper unit may be created.
(140) According to the present embodiment, it is possible to provide the image forming apparatus that can increase the rigidity in its part downstream from the discharge port in the sheet discharge direction.
Second Embodiment
(141) Using FIG. **17**, a second embodiment will be described. Hereinafter, unless otherwise specifically stated, elements on which reference characters common to the first embodiment are put possess substantially the same configuration and function described in the first embodiment, and portions different from the first embodiment will be mainly described.
(142) In the second embodiment, the document reading apparatus **300** and the document conveyance apparatus **400** are mounted on the top of the casing **100**A of the printer body **100**. The document reading apparatus **300** is an example of the upper unit that is arranged above the casing **100**A.
(143) FIG. **17** is a perspective view illustrating a frame configuration of an image forming apparatus **1**A of the second embodiment. In the second embodiment, the extension member right **148** and an apparatus supporting member right **512** are disposed on the right side plate **140**, and a column member **510** is disposed on the extension member right **148**. Further, the extension member left **150** and the apparatus supporting member left **151** are disposed on the left side plate **141**, and a column member **511** is disposed on the extension member left **150** and the apparatus supporting member left **151**. The document reading apparatus **300** (refer to FIG. **1**) is installed on upper surfaces of the column members **510** and **511**.
(144) As describe above, the extension member right **148**, the extension member left **150**, the apparatus supporting member right **512**, and the apparatus supporting member left **151** are provided with the column members **510** and **511** between them and the document reading apparatus **300**. Thereby, it is possible to increase a distance from the sheet discharge tray **135** (refer to FIG. **1**) of the printer body **100** to a bottom surface of the document reading apparatus **300**, and it is possible to increase the number of sheets that can be supported by the sheet discharge tray **135**.
(145) The apparatus supporting member right **512** is arranged at the back of the front upper stay **145** and in front of the extension member right **148**. The apparatus supporting member right **512** is fastened to the supporting member connection surface **145***b* of the front upper stay **145**, and is also fastened to the column member **510**. Further, the upper surface of the column member **510** and an upper surface of the apparatus supporting member right **512** are positioned substantially at the same height in a product height direction. Then, the document reading apparatus **300** is connected to the upper surfaces of the column members **510** and **511**, and the upper surface of the apparatus supporting member right **512**. Thereby, the document reading apparatus **300** and the printer body **100** are connected.
(146) According to the configuration described above, the document reading apparatus **300** is connected to the printer body via the extension member right **148**, the extension member left **150**, the apparatus supporting member left **151**, the apparatus supporting member right **512**, and the column members **510** and **511**.
(147) The apparatus supporting member right **512**, serving as the first supporting member, is disposed on a right front side of the image forming apparatus **1**A. The apparatus supporting member right **512** is connected to the upper surface of the right side plate **140** of the body frame **100**F, and is connected to the supporting member connection surface **145***b* that is the upright surface of the front upper stay **145**. Thereby, the inclination of the apparatus supporting member right **512** with respect to the right side plate **140** is suppressed. Then, the inclination of the document reading apparatus **300** itself with respect to the right side plate **140** is also suppressed by

connecting the apparatus supporting member right **512** and the document reading apparatus **300**.
(148) The apparatus supporting member left **151**, serving as the second supporting member, is disposed on a left front side of the image forming apparatus **1**A. With a configuration similar to the first embodiment, the inclination of the apparatus supporting member left **151** with respect to the left side plate **141** is suppressed. Then, the document reading apparatus **300** is connected to the apparatus supporting member left **151** via the column member **511**. Thereby, the inclination of the document reading apparatus **300** itself with respect to the left side plate **141** is also suppressed.
(149) According to the present embodiment described above, the supporting member connection surfaces **145***a* and **145***b* of the front upper stay **145** arranged on the front side of the image forming apparatus **1**A are respectively connected to the apparatus supporting member left **151** and the apparatus supporting member right **512**. That is, the front upper stay **145** (frame member) that is arranged downstream of the discharge port **100***d* in the sheet discharge direction D**1** includes the third surface (supporting member connection surface **145***b*) that is connected to the apparatus supporting member right **512**, and the fourth surface (supporting member connection surface **145***a*) that is connected to the apparatus supporting member left **151**. Therefore, it becomes possible to increase the rigidity of the image forming apparatus **1**A in its downstream part in the sheet discharge direction.
(150) According to the present embodiment, it is possible to provide the image forming apparatus that can increase the rigidity in its part downstream of the discharge port in the sheet discharge direction.
Modified Embodiments
(151) While, in each embodiment described above, as examples of the upper unit, the sheet processing apparatus **200** and the document reading apparatus **300** are exemplified, the upper unit may be any other unit.
(152) Further, in the first embodiment, the image forming apparatus **1** including the sheet processing apparatus **200**, the document reading apparatus **300**, and the document conveyance apparatus **400** is exemplified, and the image forming apparatus **2**, serving as the single-function printer, is exemplified as the variant example of the first embodiment. As a further variant example of the first embodiment, the image forming apparatus may be a single-function printer which includes the extension member right **148** and the extension member left **150**, but does not include the sheet processing apparatus **200**, the document reading apparatus **300**, and the document conveyance apparatus **400**.
(153) The image forming portion **100**B described above is an example of the image forming portion. The image forming portion may be, for example, an electrophotographic mechanism of an intermediate transfer system that transfers the toner image formed on the photosensitive drum **102** onto the sheet via an intermediate transfer member. The image forming portion may be a mechanism that forms a color image using developers of a plurality of colors. Further, the image forming portion may utilize mechanisms other than the electrophotographic method, such as inkjet or offset printing methods.
(154) As described above, according to the present disclosure, it is possible to provide the image forming apparatus that can increase the rigidity in its part downstream of the discharge port in the sheet discharge direction.
Other Embodiments
(155) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(156) This application claims the benefit of Japanese Patent Application No. 2023-075874, filed on May 1, 2023, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image forming apparatus comprising: a casing configured to accommodate an image forming portion that is configured to form an image on a sheet, the casing including a discharge port through which the sheet on which the image has been formed by the image forming portion is discharged in a sheet discharge direction; an upper unit arranged above the casing; a first supporting member supporting the upper unit; and a second supporting member supporting the upper unit, wherein the casing includes: a first side plate extending in a direction intersecting with a sheet width direction perpendicular to the sheet discharge direction, a second side plate extending in a direction intersecting with the sheet width direction, and a frame member arranged downstream of the discharge port in the sheet discharge direction and configured to connect the first side plate and the second side plate, wherein, in the sheet width direction, the first side plate and the first supporting member are arranged on a first side with respect to a center of the casing, and the second side plate and the second supporting member are arranged on a second side opposite to the first side with respect to the center of the casing, and wherein the frame member includes a first surface connected to the first side plate, a second surface connected to the second side plate, a third surface connected to the first supporting member, and a fourth surface connected to the second supporting member.
2. The image forming apparatus according to claim 1, wherein the casing includes a supporting portion that is configured to support the sheet discharged from the discharge port, and wherein the frame member is arranged at a position at which a distance to a downstream end of the supporting portion in the sheet discharge direction is shorter than a distance to an upstream end of the supporting portion in the sheet discharge direction.
3. The image forming apparatus according to claim 1, wherein the first surface is connected to an upper surface of the first side plate, wherein the second surface is connected to an upper surface of the second side plate, wherein the third surface is connected to the first supporting member at a position above the upper surface of the first side plate, and wherein the fourth surface is connected to the second supporting member at a position above the upper surface of the second side plate.
4. The image forming apparatus according to claim 3, wherein the first surface, the second surface, the third surface, and the fourth surface are formed by a single metal plate, and wherein the metal plate is bent between the first surface and the third surface, and is bent between the second surface and the third surface.
5. The image forming apparatus according to claim 4, wherein the frame member includes: a fifth surface that extends in the sheet width direction and is configured to connect the first surface and the second surface, and a sixth surface that extends in the sheet width direction and is configured to connect the third surface and the fourth surface, and wherein, when viewed in the sheet width direction, the metal plate is bent between the fifth surface and the sixth surface.
6. The image forming apparatus according to claim 1, wherein the frame member includes an intermediate portion extending in the sheet width direction and connecting the first surface and the second surface, and wherein at least one of an upper end of the third surface and an upper end of the fourth surface is positioned above an upper end of the intermediate portion.
7. The image forming apparatus according to claim 1, wherein the frame member includes an intermediate portion extending in the sheet width direction and connecting the first surface and the second surface, wherein the casing includes a supporting portion that is configured to support the sheet discharged from the discharge port, and wherein, when viewed in the sheet width direction, an upper end of the intermediate portion is positioned above an upper surface of the first side plate and an upper surface of the second side plate, and is positioned below the supporting portion.
8. The image forming apparatus according to claim 1, wherein a thickness of the frame member is larger than a thickness of the first side plate and a thickness of the second side plate.

9. The image forming apparatus according to claim 1, further comprising: an image processing unit that is configured to process image information, wherein the casing further includes a case member to which the image processing unit is attached, wherein part of the case member projects above an upper surface of the second side plate, and wherein the case member is connected to each of the second side plate and the second supporting member.
10. The image forming apparatus according to claim 9, wherein the casing includes: a first extension member that is arranged above the first side plate and is connected to the first side plate, and a second extension member that is arranged above the second side plate and is connected to the second side plate, and wherein the case member is further connected to the second extension member.
11. The image forming apparatus according to claim 1, wherein the casing includes: a first extension member that is arranged above the first side plate and is connected to the first side plate, and a second extension member that is arranged above the second side plate and is connected to the second side plate, and wherein an upstream part of the upper unit in the sheet discharge direction is supported by the first extension member and the second extension member, and a downstream part of the upper unit in the sheet discharge direction is supported by the first supporting member and the second supporting member.
12. The image forming apparatus according to claim 1, wherein at least one of the first supporting member and the second supporting member is connected to an upper surface of the upper unit.
13. The image forming apparatus according to claim 1, wherein the upper unit is a sheet processing apparatus that is configured to receive the sheet, on which the image has been formed by the image forming portion, from the casing and perform processing on the sheet.
14. The image forming apparatus according to claim 1, wherein the upper unit is a document reading apparatus that is configured to read image information from a document, and wherein the image forming apparatus is configured to form the image on the sheet based on the image information that the document reading apparatus has read.